

Animal Kingdom — Organised Study Notes

PU-I Biology • Teacher-issued access material

Basis of Classification

Key ideas: Levels of organisation • Symmetry • Germ layers • Coelom • Segmentation • Notochord
Write the diagnostic feature, an example and one common confusion for this group.

Body Symmetry, Germ Layers and Coelom

Key ideas: Radial • Bilateral • Asymmetry • Diploblastic • Triploblastic •
Coelomate/Pseudocoelomate/Acoelomate
Write the diagnostic feature, an example and one common confusion for this group.

Segmentation and Notochord

Key ideas: Metamerism • Chordates • Non-chordates
Write the diagnostic feature, an example and one common confusion for this group.

Phylum Porifera

Key ideas: Sponges • Cellular level • Canal system • Ostia/Spongocoel/Osculum
Write the diagnostic feature, an example and one common confusion for this group.

Phylum Coelenterata (Cnidaria)

Key ideas: Cnidoblasts • Polyp and medusa • Metagenesis
Write the diagnostic feature, an example and one common confusion for this group.

Phylum Ctenophora

Key ideas: Comb jellies • Bioluminescence • Comb plates
Write the diagnostic feature, an example and one common confusion for this group.

Phylum Platyhelminthes

Key ideas: Flatworms • Flame cells • Hermaphrodite
Write the diagnostic feature, an example and one common confusion for this group.

Phylum Aschelminthes

Key ideas: Round worms • Pseudocoelomates • Dioecious
Write the diagnostic feature, an example and one common confusion for this group.

Phylum Annelida

Key ideas: Segmented worms • Nephridia • Closed circulation
Write the diagnostic feature, an example and one common confusion for this group.

Phylum Arthropoda

Key ideas: Largest phylum • Jointed appendages • Open circulation • Malpighian tubules
Write the diagnostic feature, an example and one common confusion for this group.

Phylum Mollusca

Key ideas: Second largest phylum • Mantle cavity • Radula
Write the diagnostic feature, an example and one common confusion for this group.

Phylum Echinodermata

Key ideas: Spiny-bodied • Water vascular system • Marine

Write the diagnostic feature, an example and one common confusion for this group.

Phylum Hemichordata

Key ideas: Worm-like marine animals • Proboscis, collar, trunk

Write the diagnostic feature, an example and one common confusion for this group.

Phylum Chordata — General Features and Subphyla

Key ideas: Notochord • Dorsal nerve cord • Gill slits • Post-anal tail

Write the diagnostic feature, an example and one common confusion for this group.

Class Cyclostomata

Key ideas: Jawless • Ectoparasitic • Cartilaginous

Write the diagnostic feature, an example and one common confusion for this group.

Class Chondrichthyes

Key ideas: Cartilaginous fish • Placoid scales • Poikilothermous

Write the diagnostic feature, an example and one common confusion for this group.

Class Osteichthyes

Key ideas: Bony fish • Operculum • Air bladder

Write the diagnostic feature, an example and one common confusion for this group.

Class Amphibia

Key ideas: Water and land • Cloaca • Three-chambered heart

Write the diagnostic feature, an example and one common confusion for this group.

Class Reptilia

Key ideas: Creeping/crawling • Dry cornified skin • Oviparous

Write the diagnostic feature, an example and one common confusion for this group.

Class Aves

Key ideas: Wings and flight • Four-chambered heart • Homoiothermous

Write the diagnostic feature, an example and one common confusion for this group.

Class Mammalia

Key ideas: Mammary glands • Hair • Viviparous

Write the diagnostic feature, an example and one common confusion for this group.

Summary — Salient Features of Phyla

Key ideas: Quick comparison table

Write the diagnostic feature, an example and one common confusion for this group.